



Northern Bukidnon State College

"EmergTrack: Student Safety Monitoring System with GPS Tracking and IoT integration for Emergency Alerts"

A Capstone 1 Project Proposal

Presented to

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Northern Bukidnon State College

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Information Technology

By

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July 31, 2025



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CHAPTER 1

THE RATIONALE

Introduction

Student safety has become critical globally, especially for students who travel to school or live in off-campus housing options. The United States Department of Justice (2022) has identified school-related crimes, assaults, thefts, and physical assaults, harassment continue to be common, especially in urban areas. The United Nations Office on Drugs and Crime (2024) specified the increase in urban violence affecting young people and students living in some of the cities in Southeast Asia , which increases the responsibility to develop, maintain, and support systems, including the use of technological opportunities to meet the safety needs. According to UNICEF (2023), many students feel unsafe commuting to school because of limited or no strategies or systems aimed at creating safe commutes, especially in the developing world. Gyang and Pam (2022) concluded that off-campus students living in secure areas are subject to theft (44%) or assault (32%), robbery (20%), or even rape (4%). The Straits Times (2022) examined several reports of recent robbery and sexual violence against students in Southeast Asia and increased calls for interventions and policies for safety. The same dangers exist in the Philippines. In the first month of 2024 alone, Metro Manila reported more than 2,600 crimes, many occurring near schools and transportation terminals (Eastern Police District -NCRPO, 2023; Southern Police District - NCRPO, 2024). In the



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Municipality of Manolo Fortich, Bukidnon, student safety has become a growing concern. Based on records from 2022 to 2024, various incidents involving students have been reported. Within this three-year period, a total of 36 cases of Acts of Lasciviousness, 35 cases of Rape, and 78 cases of Physical Injuries were documented. Notably, there were no recorded cases of Robbery or Kidnapping during this timeframe. These figures reflect a consistent annual incident rate of approximately 19%, indicating a recurring threat to student welfare. The persistent nature of these cases highlights the urgent need for proactive measures to ensure student safety, especially in educational environments and during commutes to and from school Manolo Fortich MPS (2025). Paje (2025) conducted a geospatial crime analysis in Albay and demonstrated how students are victims of theft, assault, and harassment, with those risks heightened in the context of poor infrastructure and a lack of suitable emergency response systems. The Commission on Human Rights (2023), like Paje (2025), expressed concern regarding the daily threats students face, especially those associated with unsupervised commuting. Notably, Rappler (2024) revealed an increase in kidnapping events occurring through school zones. These local and international results highlighted the need for a responsive solution to student safety, such as EmergTrack—a GPS and Internet of Things, (IoT) enabled solution aimed at quickening emergency responses and parental alerts in real-time.



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In line with technological advancements, several international studies have explored the integration of IoT in emergency detection and response. Berenice S. et al. (2024) proposed an IoT-based system that uses sound recognition to detect distress screams and instantly notify a central unit. Similarly, T. Thamaraimanalan et al. (2023) developed an IoT-based safety gadget for child monitoring and notification, while Sridharan D. et al. (2025) introduced a smart wearable IoT device for child safety and real-time location tracking. In contrast, unlike wearable safety devices, EmergeTrack provides a non-wearable solution, ensuring consistent accessibility even when students lose personal gadgets. It is also a hybrid-capable student safety system that can function both offline and online. One of its most important features is that it can send emergency alerts via SMS. When the device is connected to WiFi, it can quickly send alerts to parents and police using a mobile application. It's still reliable even in places with weak signals like remote areas of Bukidnon. Locally, the LGU Manolo Fortich has implemented an emergency response mobile application using an android, Android smartphone to ask for rescue assistance during incidents regarding vehicular and other emergencies, to immediately respond to any emergencies and inquiries. As posted on the LGU Manolo Fortich Facebook Page last January 31, 2024. On July 7, 2025, the government implemented the “5-minute response time strategy”. According to Torre, “Our police officers are now visible on the streets, either on foot or in patrol vehicles equipped with radios and body cameras, ready to respond



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promptly to citizens' needs. Manolo Fortich, Bukidnon, was selected as the testing ground for the IoT-based SOS device due to its combination of rural and semi-urban characteristics, making it an ideal setting to assess the device's effectiveness. Several of its remote barangays continue to experience weak mobile network coverage and slow emergency response times, especially in mountainous and forested zones. Additionally, many students in Manolo Fortich commute long distances to school, often traversing isolated or hazardous terrain. These conditions make it a practical location to test a GPS- and SMS-based emergency alert system designed for areas where stable internet connectivity is limited and immediate assistance may not always be accessible.

To address increasing safety concerns among students, the researcher proposes EmergTrack: a non-wearable, hybrid-capable student safety system. Unlike the other systems that need smartphones or wearable gadgets like smartwatches or bracelets, EmergTrack is non-wearable yet portable; students can easily carry it in their bags or pockets. This makes it more convenient, less distracting, and always ready to use in emergencies such as crime like kidnapping, rape, theft, and robbery, with a built-in emergency button powered by GPS and GSM modules for instant location tracking and alert transmission. The EmergTrack system is built using a NODE-MCU microcontroller, a NEO-6M GPS module, and a SIM800L GSM module to help with real-time emergency responses. These coordinates are sent either through an HTTP request over WiFi or as an SMS alert via the SIM800L module, depending on



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connectivity. The data is sent to a local host website and the Blynk Cloud, so that the parents can get instant push notifications through the Blynk mobile app. At the same time, the alert data is made accessible to local authorities such as the police, which allows them to track the student's real-time location and respond quickly. By using GPS, GSM, WiFi, and cloud services in one compact system, EmergTrack gives a reliable way to send emergency alerts. It allows families and officials to work together quickly during emergencies.

1.2 Statement of the Problem

Many students are at risk from dangerous situations, particularly when they are unaccompanied or while commuting. These dangers include crimes such as harassment, theft, and physical assault. Of course, once students find themselves in danger, without immediate help, they may be unable to defend themselves or call for help promptly, on time. One limitation is that there are existing systems, but they are not real-time solutions or response systems for times when emergencies occur outside of school property. Another limitation is not having access to mobile phones in an emergency. Distraught and confused, students may panic and fail to use their mobile phones in situations that demand immediate action, and some students may not have their mobile phone available, or their mobile phone is out of battery or has no signal. While some students are at great risk, they may not be able to contact someone to help them, unless they have had enough time to call. Moreover, some students may not have access to the internet on a consistent or reliable basis. Students in rural



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communities or those facing socioeconomic challenges may not be able to depend on mobile apps or others as their means of safety alerts or emergency communication. Therefore, those particular students are left out of the safety solutions based on the internet.

1.3 Objectives of the Study

This study aims to create and develop a non-wearable, IoT-based alert system prototype called “EmergTrack” that allows students to send SMS alerts to parents with real-time GPS tracking in emergencies.

- To Conduct Planning and Data Gathering

Conduct comprehensive research to gather data about the crimes and issues related to the safety of the students. This includes the relevant crimes such as rape, kidnapping, and robbery/theft. Additionally, we gather some information about some existing IOT devices that are related to our study, that has a big impact nowadays in our society.

- To Develop and Implement

Design, create, and develop an IOT device (prototype) that enables users or students to send an SMS alert to their parents or guardians with their accurate, real-time GPS location in just one click. The development process will ensure an accurate sending of SMS alerts and also accurate GPS tracking to apply the planning and designing phase into one functional device.



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- To Test and Validate

Perform extensive testing to ensure the device will perform accurately based on the designed pattern and within the time frame. A pilot test will be conducted together with the survey for user feedback on the device (prototype).

- To Consolidate Reports and Evaluation

Compile and analyze the collected data, user feedback, and testing results into a comprehensive report. Evaluate the accuracy, effectiveness, and user satisfaction to provide recommendations for future development.

1.4 Scope and Limitations of the Study

The focus of this study is to create a simple emergency device called "EmergTrack: Student Safety Monitoring System with GPS Tracking and IoT Integration for Emergency Alerts for Parental Notification" that can help ensure the safety of students, especially minors who often travel alone. The feature of this device is that it sends an emergency alert message alongside the current GPS location of the user to parents or guardians through SMS with just one press of the button. The best part of this device is that it is so usable even without the internet; therefore, it would be highly useful in areas with an unpredictable or no connection at all. The research will cover the design, development, and initial testing of a working prototype. It will be tested in a small setting, particularly in Manolo Fortich, Bukidnon, where there are safety



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concerns involving students. This project aims to provide a more user-friendly and reliable alternative to mobile applications or wearable safety tools, especially for young users who may panic or have trouble using a phone during emergencies. The study covers technical limitations, primarily the standard SMS character limit of 160 characters under GSM encoding; this limitation further mandates that the emergency alert containing the GPS location detail must be concise so that it may be sent successfully in a single SMS. The module must also be GSM-based, such as SIM800L, which operates on 2G networks and interfaces with microcontrollers such as Arduino or ESP32. These modules were favored due to their functionality in the absence of internet connectivity, particularly useful in low-signal or offline settings.

EmergTrack is a gadget that, although helpful, cannot perform its task completely in all situations. First of all, it is not a wearable gadget, so students have to remember to bring it along with them; if forgotten, the gadget becomes functionally ineffective during an emergency. Second, as with most mobile phones that use SMS, it can only send text messages; it cannot send voice, pictures, or other media, which could otherwise be more useful during a time of crisis. GPS capabilities of the device also have their limitations: GPS signals can be lost or incorrect in specific environments, like in close proximity to high-rise buildings, during bad weather, or in low-coverage areas. In addition, the device most likely works with mobile network signals; if the area is not covered by signals, the emergency alert will not be possible. The second



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limitation is the issue of false alert; accidental activation of the emergency button might cause unreasonable alarm or response if capable protections are not in place. Finally, the device uses a battery-operated source of power, which brings with it power limitations. If the device does not get charged, it might not work at an essential time, making it useless.

1.5 Significance of the Study

This study addresses the growing concern for child safety in an age where dangers such as rape, robbery, murder, and kidnapping are becoming more common. The system introduces a technological solution that empowers children to instantly alert their parents in emergencies through a simple, one-press SOS device.

- For Parents – will benefit from faster and accurate information in emergencies, allowing a quicker response and intervention.
- For Secondary School Institution - Support student safety by providing a quick way monitor and respond when a child is in trouble.
- For NBSC RDID – By developing a reliable emergency alert system that works effectively even in low-connectivity, remote areas, it not only provides crucial assistance during critical incidents but also demonstrates NBSC's deeper commitment to protecting life and enhancing safety in line with its "Buhay" goal.
- For Future Researcher- They can use this study to create better safety devices in the future.



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1.6 Definition of Terms

- **IoT (Internet of Things)** – A network of physical devices embedded with sensors and software that allows them to connect and exchange data, typically over the internet.
 - **GPS (Global Positioning System)** – A satellite-based navigation system used to determine the exact location of a person or object anywhere on Earth.
 - **SOS Alert** – A distress signal sent during emergencies, used to notify guardians or authorities for immediate help.
 - **SMS (Short Message Service)** – A text messaging service that allows short text messages to be sent over mobile networks.
 - **Prototype** – A preliminary version of a device developed for testing and demonstration before final production.
 - **Emergency Alert System** – A mechanism or technology designed to quickly notify individuals or authorities about emergencies.
 - **Wearable Devices** – Electronic gadgets worn on the body, often used for health monitoring or personal safety.
- Non-Wearable Device** – A portable safety device that is not worn but carried and used externally, like the EmergTrack system.
- **Real-time Monitoring** – The ability to track or observe data (like GPS location) instantly as events happen.



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Panic Button – A button designed to be pressed quickly in case of an emergency to trigger an alert or call for help.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

The purpose of this review of related literature is to explore and examine existing studies, technological frameworks, and systems relevant to student safety. It focuses on the integration of Internet of Things (IoT) technology, GPS tracking, and emergency alert systems to create an effective monitoring mechanism for students. To achieve this goal, the researcher will review scholarly literature, analyze existing technological solutions, and evaluate



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theoretical foundations to identify strengths, limitations, and research gaps that the proposed system, *EmergTrack*, aims to address.

2.1 Related Literature and Studies

In urgent situations, students can be especially at risk when there's no fast way to call for help. To solve this, Mallapur et al. (2025) created a simple wearable device that lets users send their location through GPS and GSM, even if they don't have internet or a phone on hand. Similarly, Bhardwaj et al. (2023) created a child-tracking system that used Arduino, along with the GPS and GSM modules, that would send alerts to guardians using panic buttons. Ali et al. (2024) suggested an IoT-enabled smart safety device using an ESP8266, including path-logging, to log movement and alert key contacts, which can be useful in the disorienting context of a campus. Tyagi et al. (2023) proposed a smart wearable that used GPS, accelerometers, and emergency buttons that designed a model that we found particularly useful for the specific needs of students' safety. According to Española & Savandal (2016), results revealed that the respondents had personally heard, experienced, and witnessed such campus crimes as kidnappings, within-campus family feuds, murders, gunshots, ambushes, boarding house ransacks, fire incidents, and thieveries. Respondents also emphasized that they were not aware of any school administration's security plans and processes, which were perceived as effective tools to maintain campus security.



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There are times when students can't reach their phones during emergencies; maybe they're panicking, maybe their phone is dead, or maybe it's just not nearby. That's why a lot of recent studies have looked into creating safety devices that work even without a mobile phone. For instance, Ananthula et al. (2022) came up with a small wearable device built with Arduino, GPS, and GSM technology. What makes it useful is that it sends out alerts automatically, without needing a smartphone to operate. Sridharan et al. (2025) focused on child safety and designed a wearable that sends out the child's location through GSM and GPS, again without any phone involved. Vasoya (2022) developed something similar, a simple button device that sends your location through a text message using a SIM800L module. This kind of solution can be especially useful in situations where a person isn't able to reach for their phone, whether because of panic or physical limitations. Singh (2025) looked into this issue as well, creating a budget-friendly emergency tracking device using similar components. His research shows that the development of an emergency alert system, without any direct involvement of a phone, does not have to be complicated or expensive. From that, Sagolsem et al. (2023) developed an SOS that sends an SOS independent of a connection to a phone. In real-life emergencies, especially when someone can't reach their device, this kind of technology could be a lifesaver. In moments when a student can't get to their phone, maybe because they're scared, hurt, or caught off guard, having a device



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like this could make a huge difference. It could be the reason help arrives on time, and that could save lives.

According to Jaggars et al. (2021), approximately 16%-19% of college students reported technology barriers (inadequate computer hardware or Internet connection) that inhibited participation in online learning. Compared to students with robust Internet access and reliable devices, college students with inadequate technology struggled more with the transition to remote learning. Lack of access among young people aged 15-24 years old, with 759 million, or 63 percent, unconnected at home. The initiative is now collaborating under the Reimagine Education initiative and in coordination with Generation Unlimited. Through its Reimagine Education initiative, UNICEF aims to address the learning crisis and transform education by giving children and young people equal access to quality digital learning. A key to achieving this is universal internet connectivity, according to a new joint report from UNICEF and the International Telecommunication Union (ITU) (2020). This study uses data from Chile (2020–21) and shows that during school closures, formal reporting of violence against children dropped significantly by about 17% for rape and up to 43% for sexual abuse. The authors estimate that thousands of cases went unreported due to disrupted reporting channels, highlighting how losing school (and thus reporting infrastructure enabled via in-person contact and likely internet/reporting platforms) makes children, especially students, more vulnerable to sexual violence and abduction. During COVID-19 lockdowns in



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the Philippines, restrictions on movement and reduced internet access meant that students could not easily report abuse. This contributed to under-reporting of rape, lasciviousness, child abuse, and exploitation cases; despite likely increases, official data showed declines simply due to a lack of channels to report incidents. In Haiti, research found that because of widespread abductions and violent attacks, nearly all female students (96%) expressed fear of kidnapping on their way to school and up to 98% at the high school level. Many opted to stay home, especially where a lack of infrastructure (electricity, safety lighting, and internet) heightened vulnerability to kidnapping, robbery, and rape. Case studies from humanitarian contexts (e.g., DRC, Colombia, Lebanon) document that school closures and lack of alternative education, alongside limited or absent internet access, led to rises in kidnapping, sexual abuse, child labor, and recruitment into armed groups. Without connectivity to protective networks or digital reporting channels, students were left isolated and at greater risk of physical harm.

Liwanag et al. (2021) state that crime is pervasive in the Philippines and remains a major national concern, reflecting the underlying poverty that is not addressed in many areas of the country. Women and children are the most vulnerable, and innocent civilians are frequently the targets of physical and psychological abuse. Liwanag et al. (2021) suggested creating a personal emergency alert device with real-time location monitoring in order to address this expanding issue. Previous prototypes by Ahmed & Al-Neami et al. (2020)



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integrate GPS, motion sensors, and heart-rate monitoring and can send automated SMS alerts to caregivers when needed. Patrakar, S. (2025) presents an integrated school bus monitoring system designed to enhance safety and efficiency in student transportation. The system uses Arduino Uno, RFID, GPS, and GSM technologies to automatically record student attendance, track the bus's real-time location, and send SMS alerts to parents as the bus approaches designated pick-up or drop-off areas. According to Badshah et al. (2023), the transition to smart education (integrating IoT and AI into the education system) is appealing, which has a concrete impact on learners' engagement, motivation, attendance, and deep learning. According to Manuel L. et al., (2024) findings, the tabulated survey produced a computed weighted mean of 4.43 with an overall verbal interpretation of "Agree," indicating that the majority of respondents thought the prototype was effective in providing precise information like location and pulse BPM that helps a person with a medical condition or encounter an emergency like being kidnapped by informing their parents or guardians. In a prior study, Yadav and Singh (2022) developed an anti-rape wearable device that provides location-based alerts through GSM and GPS when pushed. It also contains a shock circuit and sound alarm to detect the attacker and facilitate help in real time, especially for girls walking to or from school alone. Sharma et al. (2023) had come up with a very economical safety device for women, which simply had a panic button. Once that was pressed, an SMS was sent in real-time with the GPS coordinates and recorded audio during



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critical moments of distress. The device required no internet service and could work in rural or low-service areas because it was GSM-dependent. This is a viable substitute for college students and teens, who are more likely to be vulnerable to harassment or kidnapping. The device does not require an internet connection and uses GSM to send the alert, making it suitable for rural or low-barrier areas. This may provide a low-cost option for students or teenagers who experience risk of harassment or abduction.

2.2 Synthesis of the Reviewed Related Literature and Studies

The reviewed literature emphasizes the need for technology-based solutions to ensure student safety, especially during commutes. Many existing systems use IoT, GPS, and emergency alerts but still face challenges such as poor connectivity, inaccurate indoor tracking, and delayed response times. Cases of crime affecting students, particularly in regions like Mindanao, further support the need for real-time monitoring tools. Technologies like smartwatches, school bus trackers, and wearable devices have been introduced but often rely on stable internet or power sources, which are not always available. Researchers recommend using SMS-based alerts alongside GPS to improve reliability. These insights have guided the design of the *EmergTrack* prototype, an IoT-based student safety system that features offline-capable SMS alerts, GPS, and a user-friendly interface. It aims to address the gaps in existing solutions and offer a more reliable and accessible approach to student safety.



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Table 1: **Patented Research Matrix**

Patent Research Matrix	Real-Time GPS	SMS Alerts	Non-Wearable	One-Click Activation	Portable & Battery-Powered	Web Application	Coefficient & scalable
EmergTrack: Student Safety Monitoring System with GPS Tracking and IoT integration for Emergency Alerts for Parental Notification	✓	✓	✓	✓	✓	✓	✓
Child Safety Device With GPS Tracking And Alert Messaging	✓	✓	✓			✓	✓
Advanced child safety device with GPS tracking and alert messaging	✓	✓			✓	✓	
IoT-based child tracking using RFID and GPS	✓		✓			✓	✓
IOT based Social Safety Android Application using Geolocation for Real Time Tracing	✓		✓			✓	✓
SafeGuard Bangle: A Smart Wearable with Emergency Panic Button for Women's Safety	✓	✓		✓	✓	✓	
IoT-based system for campus community security			✓				
IOT Based Smart School Bus and Student Monitoring System	✓		✓				
SafeBand: IoT-Based Smart Security Band with Instant SOS Messaging		✓			✓		
Wearable Devices and IoT	✓				✓		
Smart Mobile-Based Emergency Management and Notification System	✓	✓					✓

Figure 1 shows comparative visuals of various IoT-based safety monitoring systems and devices that have features of GPS tracking, SMS emergency alerts, and wearable features. Upon comparing the studies, systems, and devices, the “EmergTrack: Student Safety Monitoring System with GPS Tracking and IoT Integration for Emergency Alerts for Parental Notification” clearly shows a more comprehensive solution. This matrix will highlight the existing systems that may cover selected features, but none of them provide the more complete range of capabilities that EmergTrack can offer. EmergTrack will reinforce the necessity and innovation behind its development.



CHAPTER 3

METHODOLOGY

3.1 Research Design

3.1.1 Type of Research

This research study used a developmental research approach that aims to design, create, and evaluate an IoT-based device for emergency alert called “EmergTrack”. Developmental research, as opposed to simple instructional development, has been defined as the systematic study of designing, developing, and evaluating instructional programs, processes, and products that must meet criteria of internal consistency and effectiveness (Richey, 1994). This study is using an Agile developmental approach, which takes the stages of Planning, Iterative Prototyping, Implementation, Testing, and Evaluation.

To ensure a better understanding of the system’s usability and effectiveness, the study also applies the mixed-method approach, combining both quantitative and qualitative data. Quantitative data are based on the gathered responses from users through surveys and performance metrics. Meanwhile, Qualitative data are collected through interviews, observations, and feedback from the target clients, such as students, parents, or police, during testing. This combined approach and methods can provide both objective

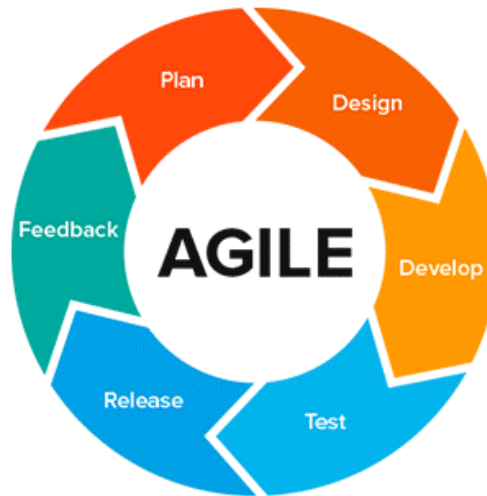


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measurements and support more in-depth insights, evaluations, and future recommendations.

3.2 Research Design Approach

In this research study, the researchers used an AGILE Developmental Approach to help this team manage to work more efficiently and more effectively while creating the highest level of device within a time frame and budget constraints. This method will follow six Fundamental phases, namely: Planning, Design, Development, Testing, and Evaluation.



Source: ("EBY, 2017)

Figure 1. **Agile Development**

In Figure 1, the researchers applied the AGILE Developmental Approach to ensure a flexible, iterative, and user-centered process throughout the system's creation. By breaking down the project into phases—Planning, Design,



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Development, Testing, Release, and Feedback—the team was able to adapt quickly to technical challenges and user needs. During planning, safety concerns and system requirements were identified; in design, the architecture involving NodeMCU, GPS, and GSM was mapped out; development involved programming and integrating modules using Arduino IDE; testing ensured functionality in real-world scenarios, including areas with weak signals; and release allowed for early deployment of a working prototype. Feedback from users like students, parents, and police helped refine the system’s emergency alert performance and usability. This approach ensured that EmergTrack was developed efficiently, met its safety objectives, and remained responsive to the evolving requirements of student protection..

Planning

The planning phase was the starting point of the project. In this stage, the researchers prepared all the needed documents and clearly described what the project aimed to do. They first analyzed the problem and gathered information to understand the needs of the community. To collect important data, the researchers made a request letter and conducted interviews and surveys with selected people. They also reviewed related studies to learn from previous research and improve their project idea. The team created the study method, set the goals, and decided how they would collect and analyze data. They made sure to get proper approvals to begin the study. To stay organized, the researchers used a Gantt chart. This chart showed the project schedule,



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divided into phases like Planning, Design, and Development. It helped them track deadlines, assign tasks, and see which activities were done or still in progress. This chart helped the team stay focused and prepared before starting the actual system development.

Design

In this phase, we make a schematic diagram and System Architecture of the device. During the first interaction, we proposed this study to our Program Head, and after it was approved, we immediately proposed it to our adviser. The entire team then discussed our approach could be to achieve our goals and objectives, and what the proposed tools to achieve the best result. The schematic Diagram, System Architecture, and Use Case Diagram were some tools used to guide the developer of what could be the flow and the overall structure of the device in this project.

3.3 Hardware and Software Specification

The following are the hardware and software specifications of the project. These were the tools required for the development of the device “EmergTrack”.

3.3.1 Hardware Specification for Development

The following are the hardware and software specifications of the project. These were the tools required for the development of the device “EmergTrack”.

Table 1. **Hardware Development Specification**

HARDWARE	SPECIFICATION
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DELL LATITUDE



- Device Name

DESKTOP-A98CDUU

- Processor

Intel(R) Core(TM) i5-7300U

CPU @ 2.60GHz 2.70 GHz

- Installed RAM 8.00

GB (7.88 GB usable)

- Graphics Card

Intel(R) HD Graphics 620

(128 MB)

- Device ID

09029F7A-8713-410B-A7F7-

0DA6D89E3DFE

- Product ID

00330-50130-52613-AAOE

M

- System Type 64-bit

operating system, x64-based

processor



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	• Pen and touch: No pen or touch input is available for this display
	• The ESP8266 is a low-cost Wi-Fi microchip used for connecting devices to the internet, enabling IoT (Internet of Things) applications. It supports both TCP/IP and microcontroller functions, allowing devices to communicate wirelessly. Commonly used in home automation, it's popular for projects that require remote monitoring and control capabilities. Manufacturer Espressif Systems • Type 32-bit microcontroller



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	• CPU Tensilica Diamond Standard 106Micro (aka. L106) @ 80 MHz (default) or 160 MHz • Memory 32 KiB instruction, 80 KiB user data • Input 16 GPIO pins • Power 3.3 V DC
	This allows the system to send emergency SMS alerts and make calls to guardians or authorities even without internet access. • NEO-6M GPS • Accuracy: ~2.5 meters • Start Time: <1s (hot), <38s (cold) • Baud Rate: 9600 bps • Voltage: 3.3V–5V



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	• Output: NMEA (GGA, RMC) • Interface: UART
	This allows the device to determine and send the user's real-time location coordinates during emergencies. • SIM800L • Bands: 850/900/1800/1900 MHz • Supports SMS, Voice, GPRS • Baud Rate: 9600 bps • Voltage: 3.7V–4.2V • SIM Type: Micro SIM • Interface: UART



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Provides portable power to the entire emergency system, enabling operation without external power sources.

- Type: Lithium Polymer (Li-Po)
- Nominal Voltage: 3.7V
- Capacity: 2500mAh
- Max Discharge Current: 2A (2000mA)
- Charging Voltage: 4.2V
- Standard Charge Current: 1A
- Connector: JST-PH 2.0 or bare wire



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Enable flexible and quick connections between components during prototyping and testing.

- Male-to-Male / Male-to-Female / Female-to-Female
- 22–26 AWG



Used to create strong, permanent electrical connections between components on a PCB. 30W–60W temperature-controlled

- Tip: Fine point
- Use: Lead-free solder recommended



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Custom PCB or universal protoboard for circuit mounting

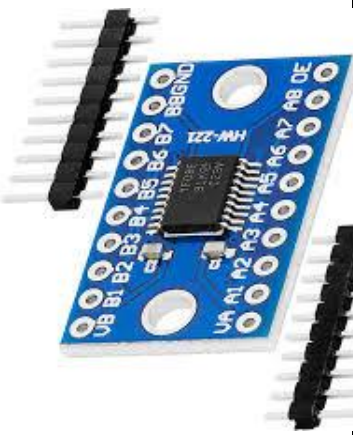
- Material: FR4



Acts as the user's emergency trigger to initiate alert functions like sending GPS location via SMS.

- Momentary tactile pushbutton
- Size: 6x6 mm common
- Contact type: Normally Open (NO)



	This allows safe communication between the 3.3V NodeMCU and 5V modules like GPS or certain sensors when programmed via the Arduino IDE. • Converts voltages (e.g., 5V ↔ 3.3V). • Typical IC: TXS0108E, BSS138-based module
---	---

3.3.2 Software Specification for Development

Table 2 discussed the software specification used to develop the system.

Table 2. Software Development Specification

SOFTWARE	SPECIFICATION
Arduino IDE	• Version: Arduino IDE 2.3.4 • Windows: Windows 10 or newer, 64-bit



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	• Linux: 64-bit AppImage or ZIP file • macOS: 64-bit Intel, 10.15: “Catalina” or newer, or 64-bit Apple Silicon, 11: “Big Sur” or newer
Xampp	• Includes Apache, MySQL (MariaDB), PHP, Perl • PHP Version: 7.4–8.2 • Apache Port: 80 • MySQL Port: 3306 • OS: Windows, macOS, Linux
VSCode	• Supported Languages: C++, Python, JavaScript, PHP, HTML, etc. • Extensions: Debugger, Git, Linters



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	- Platform: Windows, macOS, Linux - RAM: 1GB+ recommended - Disk Space: ~200MB
Leaflet	- JavaScript Mapping Library - No dependencies (lightweight) - Tile Support: OpenStreetMap, custom tiles - Core Size: ~38KB minified - Compatible with mobile & desktop browsers
Blynk	- Version: 1.2.0 Blynk Library - Supported Device: NodeMCU (ESP8266),



	SIM800L GSM module, NEO-6M GPS module • Supported Connectivity: WiFi, GSM (via SIM800L using HTTP), Blynk Cloud • Android: Android OS version 5+
--	--

3.3.3 Hardware Specification for Deployment

Table 3 discussed the hardware specification used during the project's deployment stage.

Table 3. **Hardware Deployment Specification**

HARDWARE	SPECIFICATION
Computer Desktop	Processor: Intel(R) Core(TM) i3-7020U CPU @ 2.30GHz 2.30 GHz Installed RAM: 8.00 GB System type: 64-bit operating system, x64-based processor



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	<i>**Atleast or Higher than the above-mentioned specification</i>
--	---

3.3.4 Software Specification for Deployment

Table 4 discussed the hardware specification used during the project's deployment stage.

Table 4. **Hardware Deployment Specification**

HARDWARE	SPECIFICATION
Thonny Python IDE	Python Version: Python 3.10 Minimum Tk Version: Requires Tk 8.6 File Size: Approximately 23.48 MB License: Free and open-source



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	Supported Platforms: Windows, macOS, and Linux <i>**Atleast or Higher than the above-mentioned specification</i>
--	---

3.4 System Design Architecture

The Diagram below presents the system architecture of EmergTrack: Student Safety Monitoring System with GPS Tracking and IOT integration for Emergency alerts. It will describe the workflow of the device and how the device will function in times of emergencies. This device is built using an IoT-based design architecture to ensure reliability, accuracy, and accessibility for students, especially minors who often commute alone.

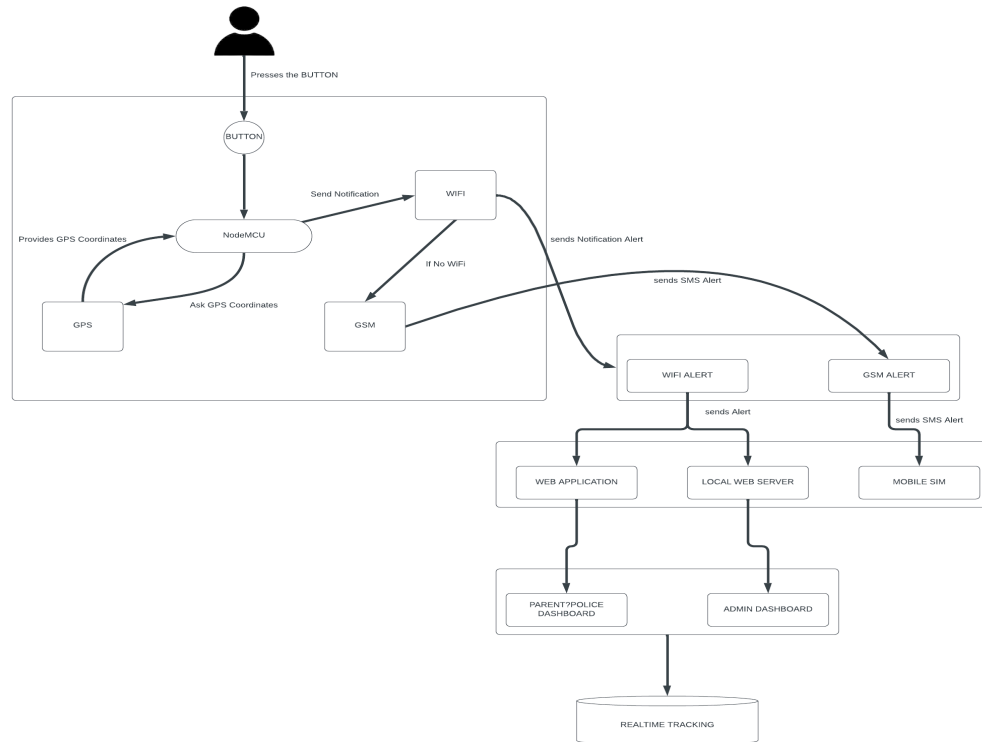


Figure 2. System Design Architecture of EmergeTrack

As shown in Figure 2, the EmergTrack system architecture illustrates how a student's emergency alert is processed and delivered in real-time. When the emergency button is pressed, the device retrieves GPS coordinates via the GPS module. It first checks for Wi-Fi availability—if available, the system sends the alert through Wi-Fi to both a local web server (admin dashboard) and a cloud-based web application (parent/police dashboard), enabling real-time tracking via embedded maps (e.g., Google Maps). If Wi-Fi is unavailable, the alert is sent via a GSM module through SMS to a mobile SIM. Annotated in the



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architecture are key components such as the Emergency Button, GPS, Wi-Fi/GSM modules, and respective dashboards, showing the flow of data and decisions made by the system. Under optimal network conditions, alerts can be sent and received within 3–5 seconds via Wi-Fi, and 5–10 seconds via GSM, depending on network signal and SMS delays.

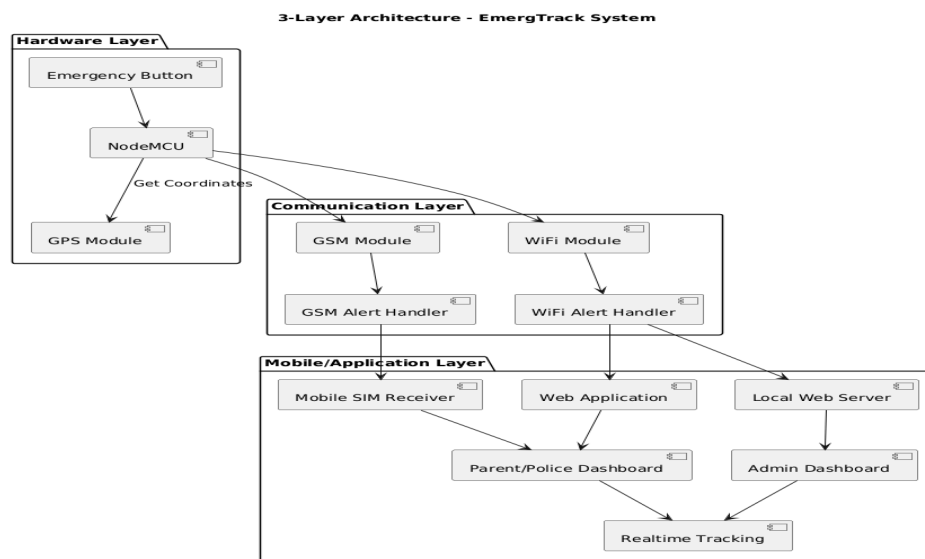


Figure 3. Layered Architecture of EmergTrack

Figure 3, shows the layered architecture of EmergTrack organizes the workflow into the Hardware layer, the Communication Layer and the Application Layer. This Structure ensures efficient alert handling and real-time monitoring across both offline and online scenarios.

3.5 System Diagrams

Use-Case Diagram



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The proponents used a Use-Case Diagram to describe what the application does and how the operators use it.

USE CASE DIAGRAM OF "EmergTrack"

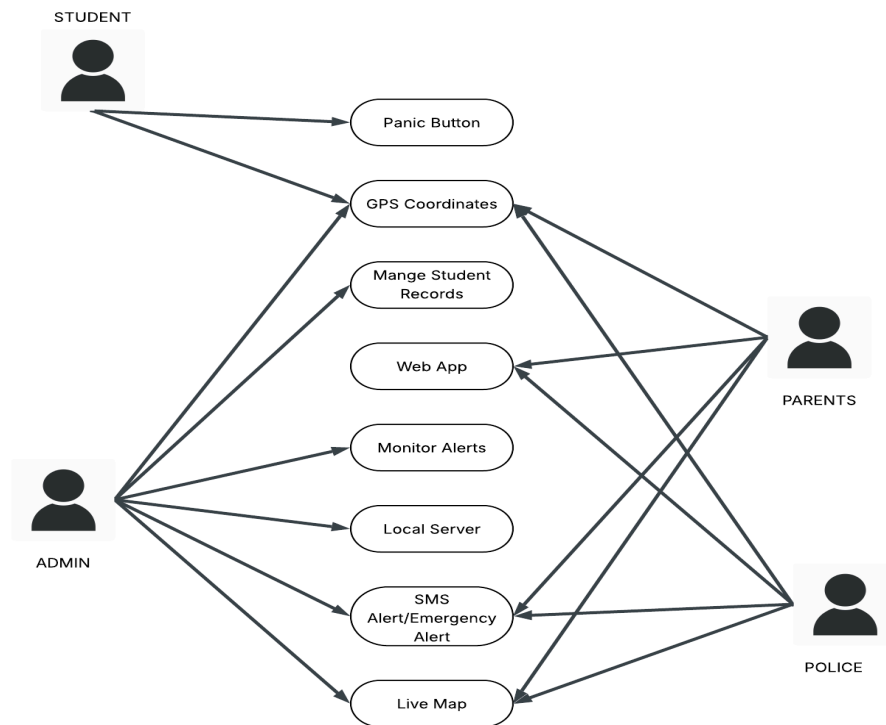


Figure 4. Use-Case Diagram of EmergTrack

Figure 4, shows the use case diagram of EmergTrack, which illustrates the emergency response and tracking system where a student will press the panic button to trigger the SMS alert with real-time GPS coordinates. These GPS coordinates are sent to the parents, admin, and police station if possible through the platforms of Blynk for graphical representation, a web application where the actors can monitor the real-time location of the student through a map, SMS



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alerts if the device is offline and does not have an internet connection. The admin can also monitor and manage the system via a local website for data analysis and effective data logging. This setup enables powerful action and location tracking during student emergencies

3.1.6 Flow Chart.



EmergTrack Flowchart

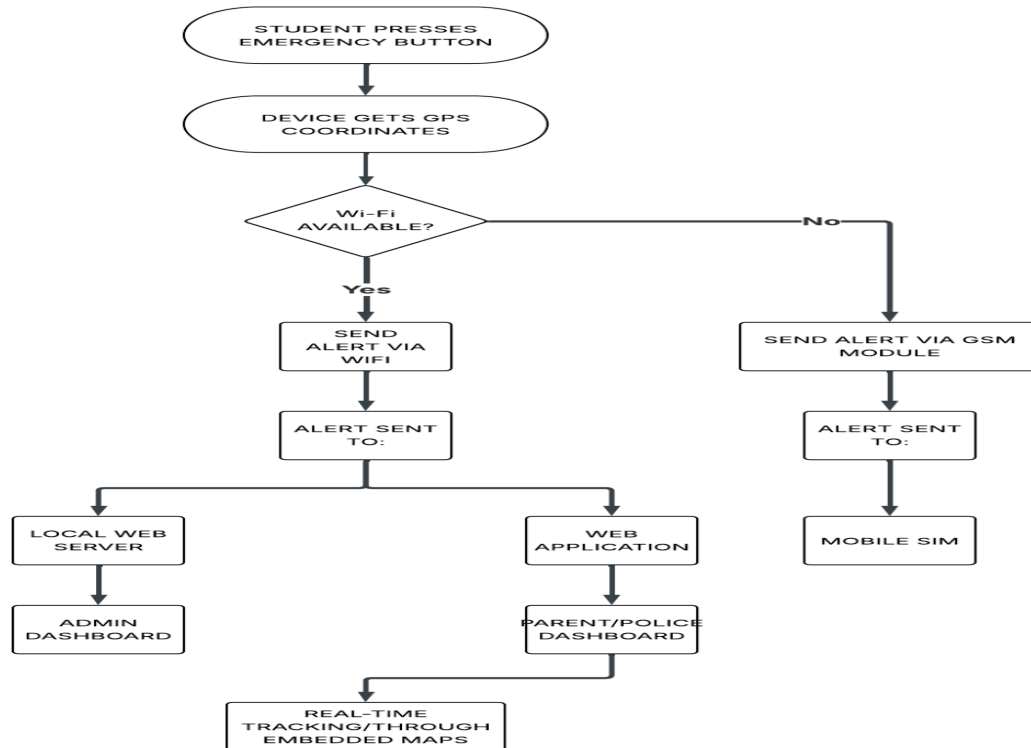


Figure 4. Flow Chart Diagram of EmergTrack

Figure 4, illustrates the System flow of “EmergTrack” where pressing the emergency button will trigger the device to request the student’s GPS location. If wifi is available, the alert is sent via wifi to a localhost web server for admin monitoring and to a web application for parents or police to track the student in real-time through embedded maps. If WiFi is unavailable, the alert is sent via GSM module through direct SMS to a mobile phone’s SIM, ensuring emergency alerts are delivered with or without internet access.

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